SE 3002 — Software Quality Engineering

FAST School of Computing — Department of Software Engineering — FAST NUCES Islamabad

ASSIGNMENT 3

Black-Box Testing: Equivalence Partitioning & Boundary Value Analysis

System Under Test: Testfolio (testfol.io)

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Course	SE 3002 — Software Quality Engineering
Technique	Black-Box Testing: EP & BVA (Manual)
Total Marks	**100 marks**
Submission	LMS — one PDF file per student

1. Background

This assignment requires you to apply Equivalence Partitioning (EP) and Boundary Value Analysis (BVA) to a real, deployed web application and produce a complete, evidence-backed test plan.

Because this is a live system, its exact input rules are not documented for you. You will discover them through careful exploration (probe testing), form your own hypotheses about valid and invalid input classes and boundaries, and design test cases accordingly. This mirrors professional test engineering practice.

2. System Under Test: Testfolio

Testfolio (https://testfol.io/) is a web-based portfolio back-testing and analysis platform. Users can:

- Build investment portfolios by specifying assets (stocks/ETFs) and their allocation percentages.
- Run historical back-tests over user-specified date ranges with a chosen initial investment amount.
- Configure rebalancing frequency, deposit schedules, and benchmark comparisons.
- Visualise portfolio performance through charts and summary statistics.

> **Note:** *Create a free account at https://testfol.io/start before beginning. All testing must be performed on your own account.*

3. Objectives

By completing this assignment you will be able to:

- Apply weak and strong EP to identify valid and invalid input classes for diverse input types.
- Apply 2-value and 3-value BVA to numeric ranges, string lengths, date constraints, and percentage boundaries.
- Write a structured test plan for a real web application.
- Document test cases with unique IDs, input values, expected results, and actual results.
- Build a traceability matrix linking each test case to an input field and technique.
- Provide execution evidence (screenshots) demonstrating that every executed test case was actually run.

4. Scope

The following eleven forms/tools are in scope for this assignment. Visit each URL, explore the interface, and identify all relevant input fields yourself — discovering the inputs is part of the task.

#	Form / Tool	URL
1	Create Account (Sign Up)	testfol.io/start
2	Sign In (Login)	testfol.io/login
3	Portfolio Backtester	testfol.io/
4	Monte Carlo Simulator	testfol.io/monte-carlo
5	Safe Withdrawal Rate Calculator	testfol.io/safe-withdrawal-rate-calculator
6	Backtest Optimizer	testfol.io/backtest-optimizer

#	Form / Tool	URL
7	Efficient Frontier	testfol.io/efficient-frontier
8	Contact Form	testfol.io/contact
9	Lump Sum vs DCA	testfol.io/lumpsum-vs-dca
10	Time Value of Money Calculator	testfol.io/time-value-of-money
11	Rebalancing Sensitivity	testfol.io/rebalancing-sensitivity

Note: *Validation rules (allowed ranges, minimum/maximum values, required-field logic) are intentionally withheld. Discover them through probe testing and record every probe in your Probe Log (TP-10).*

5. Probe Testing Guidance

Before writing test cases, spend time exploring the application. This exploration is called probe testing.

What to probe

- Enter values at obvious extremes: empty, a single character, very long string, negative numbers, zero, decimals.
- Observe error messages: does the message name the field? Does it reveal the valid range? Different messages for different violations suggest distinct equivalence classes.
- Try special characters: apostrophes, hyphens, unicode, spaces, SQL fragments.
- For date inputs: past dates, future dates, same start and end, end before start.
- For percentages: 0%, 100%, 101%, negative, decimals, values that don't sum to 100%.
- Try submitting forms with blank fields and with partial data.

Documenting your probes

Maintain a probe log (TP-10 in the template). A probe is a complete form submission — list ALL inputs you entered, not just the field under investigation. Context fields (needed to make the form submittable) must be included alongside the value being tested.

Probe ID	Form	Test Inputs (field: value, ...)	System Response	Inference
PR-001	Sign Up	Email: 'test@x.com', Password: '' (empty)	'Password is required'	Password is required; min length > 0
PR-002	Portfolio Backtester	AAPL: 60%, MSFT: 60% (total = 120%)	'Allocations must sum to 100%'	Total allocation validated; upper bound = 100%
PR-003	Monte Carlo Simulator	Start: 2099-01-01, End: 2100-01-01, Amount: 10000	Simulation ran; no error message	Future start dates are silently accepted

Note: *These are illustrative examples — do not copy them. Your test cases must arise from your own probes on your own account.*

6. Deliverables

Submit a single PDF document containing all five deliverables. The student template provided has pre-formatted sections for each.

Deliverable	Description	Marks
D1 — Test Plan	Structured test plan covering scope, approach, environment, schedule, risks, and probe log	20
D2 — EP Test Case Tables	Partition tables for every in-scope input field, with VEC/IEC classes and representative values	25
D3 — BVA Test Case Tables	Boundary tables for all applicable input fields, with boundary probing values and test cases citing BC IDs	25
D4 — Traceability Matrix	Matrix mapping every test case to its field, technique, class/boundary, and pass/fail status	10
D5 — Execution Log	Screenshots and actual results for every executed test case; defect observations	20
	TOTAL	100

D1 — Test Plan [20 marks]

Your test plan must include all of the following sections:

Section	Content required	Marks
TP-1 Metadata	Plan ID, version, author (roll number), status	1
TP-2 Introduction	Brief description of Testfolio and purpose of this plan	2
TP-3 Scope	Features in scope for this assignment	2
TP-4 Test Objectives	What the testing is intended to verify or reveal	2
TP-5 Test Approach	How you applied EP and BVA; your probe strategy	3
TP-6 Test Environment	Browser, OS, resolution, test account details	2
TP-7 Entry & Exit Criteria	Conditions for starting and stopping testing	2
TP-8 Test Schedule	Estimated time per area; activities	2
TP-9 Risks & Mitigations	At least 3 risks with mitigation strategies	2
TP-10 Probe Log (Appendix A)	Table of all probes performed during exploration (min. 15 probes)	2

D2 — Equivalence Partitioning [25 marks]

D2 has two parts. First, complete an Analysis Table for every input field you test. Then, write your EP test cases in the combined EP Test Case Table, citing the EC IDs you assigned.

Layer 1 — EP Partition Tables (one per input field)

For each input field across all eleven forms, produce one partition table that lists all its equivalence classes and assigns each a unique EC ID. Complete all partition tables before designing any test cases.

Column	What to write
EC ID	Unique equivalence class ID — EC-001, EC-002, … (sequential across all fields)
Equivalence Class Description	Describe the partition precisely (e.g. 'Valid email address format', 'Starting value below minimum')
Representative Value	A specific value that sits inside this equivalence class
Type (VEC / IEC)	Valid Equivalence Class or Invalid Equivalence Class

Layer 2 — EP Test Case Table (one combined table)

Design test cases that exercise the equivalence classes you identified. Apply the following technique tiers:

- All forms — Weak EP with Robustness: design one test case per equivalence class, including each invalid equivalence class (IEC). Record which EC IDs each test case exercises.
- Critical form only — Strong EP: for the form you designate as most complex (see note below), additionally design test cases that cover multiple equivalence classes simultaneously in a single TC.

Note: *Designate your critical form: identify the one form you consider most complex or highest-risk. State your designation and justification in TP-5 (Test Approach). That form must receive both Strong EP and 3-point BVA.*

Column	What to include
TC-EP ID	Unique test case ID — e.g. TC-EP-001
EC IDs Covered	List all equivalence class IDs exercised by this TC (e.g. EC-001, EC-017)
Feature / Form	The form being tested
Input Field(s)	The field(s) involved in this test case
Test Value(s)	The exact value(s) entered
Expected Result	What the system should do
Actual Result	What the system actually did (fill in during execution)
P/F	Pass or Fail

D3 — Boundary Value Analysis [25 marks]

D3 has two layers. Determine which input fields have meaningful boundaries — this judgment is part of the assignment. For each such field, identify its boundaries in Layer 1. Then design test cases in Layer 2 citing those BC IDs.

Layer 1 — BVA Boundary Tables (one per applicable input field)

For each field where you determine BVA applies, produce one boundary table identifying its boundaries and assigning each a unique BC ID. Discover boundary values through probe testing.

Column	What to write
BC ID	Unique boundary ID — BC-001, BC-002, … (sequential across all fields)
Feature / Form	The form this input field belongs to (e.g. 'Sign Up', 'Monte Carlo Simulator')
Input Field	The specific input field containing this boundary (e.g. 'Password', 'Rolling Window')
EC ID	The equivalence class this boundary defines — reference the EC ID from your D2 partition tables (e.g. EC-004)
Boundary Description	Describe the boundary precisely (e.g. 'Minimum password length', 'Upper bound of Rolling Window')
Boundary Value	The exact value at the boundary, inferred from probe testing

Layer 2 — BVA Test Case Table (one combined table)

Design test cases that probe each boundary. Apply the following technique tiers:

- All forms — 2-point BVA: for each boundary, test two values (e.g. just below the boundary and at the boundary, or at the boundary and just above). Each TC cites the BC ID it probes and states the position of the test value relative to the boundary.
- Critical form only — 3-point BVA: for the form designated as critical, test three values per boundary — one below, one at, and one above. All three positions must be covered for each boundary in the critical form.

Column	What to include
TC-BVA ID	Unique test case ID — e.g. TC-BVA-001
BC IDs	Boundary ID(s) being probed (e.g. BC-003)
Feature / Form	The form being tested
Input Field	The field containing the boundary
Test Value	The specific value entered
Position	Relative to boundary: Below / At / Above
Expected Result	What the system should do
Actual Result	What the system actually did (fill in during execution)
P/F	Pass or Fail

Note: *Every BC ID defined in Layer 1 must have at least one corresponding test case in Layer 2.*

D4 — Traceability Matrix [10 marks]

One row per test case (both EP and BVA). Columns: TC ID, Feature/Form, Input Field, Technique, Class/Boundary Description, Status (Executed/Not Executed), Result (Pass/Fail). Must include every test case from D2 and D3.

D5 — Execution Log [20 marks]

For every executed test case, provide the TC ID and screenshot(s) proving the test was run. A single screenshot that shows both the inputs entered and the system's response is acceptable; otherwise provide one screenshot showing the input state and one showing the response. The verdict, actual result, and defect observations are already recorded in your D2 and D3 tables — do not duplicate them here.

> ⚠ **Important:** *Screenshots must be legible — crop to show the relevant part of the screen. A full-desktop screenshot with a tiny form is not acceptable. Use a consistent browser and zoom level throughout.*

> ⚠ **Important:** *Your roll number must be visible in every screenshot. Place a physical sticky note or post-it bearing your roll number on your screen so it appears within the cropped area of each screenshot. Screenshots without a visible roll number will not be graded.*

7. Submission

File	Naming convention	Format
Completed assignment	A3_[RollNo]_[YourName].pdf	PDF only

- Submit through LMS by the posted deadline.
- Include your roll number on every page (use the header/footer in the template).
- All screenshots must be embedded in the PDF — not attached separately.
- The PDF must be readable without a password.

8. Marking Rubric

Deliverable	CLO	Criteria for Full Marks	Marks	Partial / Zero
D1 — Test Plan	CLO3	All 10 sections present; scope aligned with assignment; approach describes EP & BVA usage; ≥3 meaningful risks; ≥15 probes in log	20	Section missing = −2 each; no probe log = −4
D2 — EP Tables	CLO2	EP Partition Tables present for every in-scope field; each EC has a unique ID and correct VEC/IEC classification. EP Test Case Table: Weak EP with Robustness applied to all forms (one TC per EC, including IECs); Strong EP applied to the designated critical form; EC IDs cited in every TC	25	Missing fields = −2 each; EC IDs absent = −1 per TC; VEC/IEC confused = −0.5 per class; Robustness (IEC coverage) missing = −5; Strong EP absent for critical form = −5
D3 — BVA Tables	CLO2	BVA Boundary Tables present for all applicable fields; 2-point BVA applied to all forms; 3-point BVA applied to the designated critical form; BC IDs cited in every TC; position (Below/At/Above) correctly stated; judgment shown in field selection for BVA	25	BC IDs absent = −1 per TC; position column missing = −5; 3-point BVA absent for critical form = −5; boundaries not probed = max 15
D4 — Traceability Matrix	CLO3	All TCs listed; columns complete; status and result match execution log; no missing TCs	10	TCs missing = −1 each; status mismatches = −0.5 each
D5 — Execution Log	CLO2	TC ID cited for every executed test case; legible screenshot(s) present showing both input entered and system response; all in-scope input fields have at least one executed TC with evidence	20	TC ID missing = −0.5 each; illegible screenshots = −1 each; fields with no executed TC evidence = −2 each
TOTAL				**100**

9. Academic Integrity

⚠ **Important:** *Each student must independently probe the application, design their own test cases, and produce their own execution screenshots. Shared or copied test cases receive zero marks for all students involved.*

- Your probe log must contain your own probes, run on your own account.
- Your test cases must reflect your own observations of the system.
- Your screenshots must show your own browser session with your roll number visible on a sticky note on the screen.
- AI-generated test cases without matching execution evidence receive zero marks. The viva will probe your understanding of each test case you submitted.

ℹ **Note:** *Grading is viva-based. You may be asked to explain any test case, re-run it live, or demonstrate the behaviour you documented. Ensure you understand everything you submit.*

10. Frequently Asked Questions

Q: The application doesn't show error messages for some invalid inputs. What should I do?

Document the behaviour you observe (e.g. 'the form clears and reloads' or 'the input is silently ignored'). If the system silently accepts an invalid value, that may itself be a defect observation.

Q: I can't find the exact minimum or maximum value for a field. How many probes are enough?

Use binary search: if 10 is accepted and 1,000 is rejected, try 100, then 50, etc. You don't need the exact digit, but your BVA values must bracket the boundary you infer.

Q: Can I use browser developer tools?

You may use DevTools to observe network responses. You may not use client-side source code to infer rules — this is a black-box exercise.

Q: Do I need to test on multiple browsers?

No. Choose one browser (Chrome or Firefox recommended) and use it consistently throughout. Note the browser and version in your test environment section.

— End of Assignment 3 Handout —